An exact anti-degradability threshold for the three-level bosonic loss–dephasing channel

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Abstract

We determine the exact anti-degradability threshold of the bosonic loss–dephasing channel restricted to its first three Fock levels, at Gaussian phase variance $\ln 2$. The threshold is an isolated root of a sextic polynomial, approximately 0.664544268033. A rank-four symmetric Choi extension establishes feasibility at the endpoint. Explicit separating operators, lifted from the population support to the full extension space, exclude every larger transmissivity. Exact loss composition covers the remaining parameter intervals. This certifies a previously numerical boundary and sharpens a necessary condition for anti-degradability of the unrestricted bosonic channel.

Keywords: bosonic quantum channel; photon loss; Gaussian dephasing; anti-degradability; symmetric extension

1 Introduction

Photon loss and random phase noise act on different parts of a bosonic state: loss changes photon-number populations, while number dephasing suppresses coherences. Their combined channel can be anti-degradable even in a parameter region where pure loss alone is not. An anti-degradable channel has zero unassisted quantum capacity, because its environment can simulate the receiver. Determining the boundary of this property therefore provides a useful, although incomplete, communication criterion.

Mele, Salek, Giovannetti and Lami studied the bosonic loss–dephasing channel and its finite Fock restrictions [1]. Their two-level threshold is $1/(1+e^{-\gamma})$. They also established equality with this bound for the three-level restriction when $e^{-\gamma} \leq \sqrt{2}-1$; outside that regime the three-level boundary was evaluated numerically. We fix $\gamma = \ln 2$, where that analytic hypothesis does not apply. Our contribution is the exact endpoint and explicit primal and full-space dual certificates for this previously numerical point. We do not claim that the qualitative departure from the two-level boundary was previously unknown.

The argument exploits zeros in the channel's population support and its number-phase covariance. Swap invariance alone does not justify restricting an extension to the symmetric output subspace in arbitrary qutrit problems [2]. Here that restriction follows from a particular marginal-preserving transfer of antisymmetric blocks. A positive operator constructed on the resulting support is then lifted to the entire $3 \times 3 \times 3$ extension space. This distinction is essential for an unrestricted infeasibility certificate.

2 Channel, conventions and result

Let $\mathcal{H}_3 = \mathrm{span}\{|0\rangle, |1\rangle, |2\rangle\}$. The exact restriction of vacuum pure loss followed by centered Gaussian number dephasing is

$$\mathcal{N}_\lambda(|m\rangle\langle n|) = 2^{-(m-n)^2/2} \sum_{\ell=0}^{\min(m,n)} \sqrt{\binom{m}{\ell}\binom{n}{\ell}}\,(1-\lambda)^\ell \lambda^{(m+n)/2-\ell} |m-\ell\rangle\langle n-\ell|, \tag{1}$$

for $m, n \in \{0, 1, 2\}$ and transmissivity $0 \le \lambda \le 1$. The dephasing convention is $e^{-\gamma(m-n)^2/2}$ with phase variance $\gamma = \ln 2$, as in Lemma 25 of Ref. [1]. The parameters, phases in radians, photon numbers and all matrix entries are dimensionless. This is an invariant finite-dimensional restriction: loss only removes photons. There is no output projection, postselection or renormalization, and no perturbative or numerical truncation of the channel.

In the input-first tensor convention, define the normalized Choi state

$$J_\lambda = \frac{1}{3} \sum_{m,n=0}^{2} |m\rangle\langle n| \otimes \mathcal{N}_\lambda(|m\rangle\langle n|), \qquad \mathrm{Tr}_B\, J_\lambda = I_A/3. \tag{2}$$

Writing $|ab\rangle = |a\rangle_A \otimes |b\rangle_B$, its nonzero entries are

$$\begin{aligned} 3J_\lambda = {} & |00\rangle\langle 00| + (1-\lambda)|10\rangle\langle 10| + \lambda|11\rangle\langle 11| \\ & + (1-\lambda)^2|20\rangle\langle 20| + 2\lambda(1-\lambda)|21\rangle\langle 21| + \lambda^2|22\rangle\langle 22| \\ & + \left[\sqrt{\lambda/2}|00\rangle\langle 11| + \frac{\lambda}{4}|00\rangle\langle 22| + \sqrt{\lambda}(1-\lambda)|10\rangle\langle 21| + \frac{\lambda^{3/2}}{\sqrt{2}}|11\rangle\langle 22| + \mathrm{h.c.}\right]. \end{aligned} \tag{3}$$

No partial transpose is included. A channel is anti-degradable precisely when its Choi state has an extension X satisfying

$$X \succeq 0, \qquad SXS = X, \qquad \mathrm{Tr}_{B_2}\, X = J_\lambda, \tag{4}$$

where S exchanges B_1 and B_2 [1, 2]. Swap symmetry gives the other marginal. For completeness, an anti-degrading map applied to a Stinespring environment gives a channel with two identical output marginals; swap averaging gives (4). Conversely, the extension defines such a two-output channel. Uniqueness of Stinespring dilation expresses its second output as a channel from the environment of the first, providing an anti-degrading map.

Theorem 1. *Let r be the unique root in*

$$I_r = \left[\frac{66454426803}{10^{11}}, \frac{66454426804}{10^{11}}\right] \tag{5}$$

of

$$P(x) = 601x^6 - 714x^5 + 113x^4 - 1040x^3 + 2080x^2 - 1280x + 256. \tag{6}$$

Then $\mathcal{N}_\lambda$ is anti-degradable if and only if $\lambda \le r$. At r there is a rank-four normalized swap-invariant extension. For every $\lambda > r$ there is an explicit Hermitian H_λ such that

$$W_{H_\lambda} := \frac{H_\lambda \otimes I_3 + S(H_\lambda \otimes I_3)S}{2} \succeq 0, \qquad \mathrm{Tr}(H_\lambda J_\lambda) < 0 \tag{7}$$

on the full 27-dimensional space.

An exact narrower enclosure is

$$\frac{66454426803317}{10^{14}} < r < \frac{66454426803318}{10^{14}}. \tag{8}$$

Thus $r \simeq 0.664544268033$, slightly below the two-level bound $2/3$. The decimal is descriptive; the proof uses the algebraic definition and rational enclosures.

3 Population support and the two swap sectors

Set $Y = 3X$, so that $\mathrm{Tr}_{B_2} Y = 3J_\lambda$ and $\mathrm{Tr}\, Y = 3$. All marginal populations with $b > a$ vanish. For a positive Y, the corresponding diagonal summands $Y_{abc,abc}$ are nonnegative and hence individually zero. A zero diagonal of a positive matrix forces its whole row and column to vanish, by $|Y_{ij}|^2 \le Y_{ii}Y_{jj}$. The second marginal imposes $c \le a$ as well. Every extension therefore has support in

$$\mathcal{K} = \mathrm{span}\{|abc\rangle : b, c \le a\}, \qquad \dim \mathcal{K} = 1 + 4 + 9 = 14. \tag{9}$$

The Choi state is invariant under $e^{i\theta(-N_A+N_B)}$. To average an extension, use the representation

$$U_\theta = e^{i\theta(-N_A+N_{B_1}+N_{B_2})}. \tag{10}$$

This preserves positivity and each marginal and gives blocks of charge $q = b + c - a$. The averaging commutes with the swap. For $b < c$ define

$$|a; s_{bc}\rangle = \frac{|abc\rangle + |acb\rangle}{\sqrt{2}}, \qquad |a; a_{bc}\rangle = \frac{|abc\rangle - |acb\rangle}{\sqrt{2}}; \quad |a; s_{bb}\rangle = |abb\rangle. \tag{11}$$

The symmetric sector has charge dimensions $(1, 2, 4, 2, 1)$, and the antisymmetric sector has dimensions $(1, 2, 1)$ at $q = -1, 0, 1$.

For this support, every antisymmetric block can be transferred isometrically to the corresponding symmetric kets without changing either marginal. The singleton kets are $|2; a_{01}\rangle$ and $|2; a_{12}\rangle$. Their diagonal marginal images coincide with those of the symmetric kets. The only nonsingleton block has $|1; a_{01}\rangle$ and $|2; a_{02}\rangle$. Its two diagonal units have the same property. In its off-diagonal unit the only common traced number is zero; the two antisymmetric signs agree, so the partial trace equals that of the symmetric unit. The same holds for its adjoint and for the other partial trace. These six ordered matrix units exhaust all blocks. Adding the transferred positive matrices to the symmetric blocks preserves positivity. This proves symmetric-support equivalence for the present channel; no such equivalence for general qutrit channels is used.

Since J_λ is real, averaging with complex conjugation permits real symmetric blocks. Their ordered bases are in Table 1. Let B_q denote the isometry whose columns are the listed kets. Direct partial tracing of these kets gives the ten independent equations

Table 1: Ordered symmetric block bases in the necessary support. The scalar blocks are denoted ζ and w.

Charge	Block	Ordered basis
-2	ζ	$\|2; s_{00}\rangle$
-1	R	$\|1; s_{00}\rangle,\ \|2; s_{01}\rangle$
0	T	$\|0; s_{00}\rangle,\ \|1; s_{01}\rangle,\ \|2; s_{02}\rangle,\ \|2; s_{11}\rangle$
1	V	$\|1; s_{11}\rangle,\ \|2; s_{12}\rangle$
2	w	$\|2; s_{22}\rangle$

$$
\begin{aligned}
T_{00} &= 1, & T_{01} &= \sqrt{\lambda}, \quad T_{02} = \lambda/(2\sqrt{2}),\\
R_{00} + T_{11}/2 &= 1-\lambda, & T_{11}/2 + V_{00} &= \lambda,\\
R_{01} + T_{13} &= \sqrt{2\lambda}(1-\lambda),\\
T_{12} + \sqrt{2}V_{01} &= \sqrt{2}\lambda^{3/2},\\
\zeta + R_{11}/2 + T_{22}/2 &= (1-\lambda)^2,\\
R_{11}/2 + T_{33} + V_{11}/2 &= 2\lambda(1-\lambda),\\
T_{22}/2 + V_{11}/2 + w &= \lambda^2.
\end{aligned}
\tag{12}
$$

All other marginal entries vanish. The problem now has 18 real matrix entries, with ten independent constraints, while retaining equivalence to (4).

4 The algebraic endpoint and its extension

At the specified root r, set

$$
t = \sqrt{r}, \quad d = 2-3r, \quad p = \frac{r}{2\sqrt{2}}, \quad \alpha = 2(1-r)^2 - \frac{r^2}{8}, \quad k = \frac{t}{\sqrt{2}}. \tag{13}
$$

All square roots below are the positive real roots. Set $\zeta = 0$, $R = 0$, and

$$
G = \begin{pmatrix} 1 & 0 \\ t & \sqrt{d} \\ p & \sqrt{\alpha} \\ kt & k\sqrt{d} \end{pmatrix}, \qquad T = GG^{\mathsf{T}}, \qquad h = \begin{pmatrix} \sqrt{2r-1} \\ \sqrt{2r(1-r)} \end{pmatrix}, \qquad V = hh^{\mathsf{T}}, \quad w = r^2 + r - 1. \tag{14}
$$

The positivity of $d, \alpha, 2r-1, w$ follows from the rational interval (5); exact bounds are summarized in Appendix A. Thus these blocks are positive semidefinite. Substitution in (12) leaves only one residual equation,

$$
\sqrt{(2-3r)(2(1-r)^2 - r^2/8)} + 2\sqrt{r(1-r)(2r-1)} = \frac{3r^{3/2}}{2\sqrt{2}}. \tag{15}
$$

To check the branch after squaring, define

$$
\begin{aligned}
A(x) &= (2-3x)(16(1-x)^2 - x^2),\\
B(x) &= 32x(1-x)(2x-1), \qquad C(x) = 9x^3.
\end{aligned}
\tag{16}
$$

Direct expansion gives

$$
(C - A - B)^2 - 4AB = 4P(x). \tag{17}
$$

On I_r, $A, B, C, C-A-B$ are strictly positive. Consequently $P(r) = 0$ implies $C - A - B = 2\sqrt{AB}$ and then $\sqrt{A} + \sqrt{B} = \sqrt{C}$, which is (15). There is no extraneous negative radical branch. Sturm counts establish exactly one root in I_r and exactly one in $(3/5, 2/3)$. Applying the same exact count to (8) gives its narrower enclosure.

The endpoint extension is now explicit:

$$
X_r = \frac{1}{3}\left[B_0 GG^{\mathsf{T}} B_0^{\mathsf{T}} + B_1 hh^{\mathsf{T}} B_1^{\mathsf{T}} + w|222\rangle\langle 222|\right]. \tag{18}
$$

Equations (12)–(15) give both marginals J_r. Each component is supported on symmetric output kets. Moreover $\operatorname{Tr} T + \operatorname{Tr} V + w = 3$, so $\operatorname{Tr} X_r = 1$. The two columns of G are independent because $d > 0$; $h \neq 0$ and $w > 0$. The three charge sectors are orthogonal, proving $\operatorname{rank} X_r = 4$.

5 A separating operator on the full population support

The following coefficients are fixed at r, even when the tested transmissivity is varied. Define

$$u = \sqrt{\alpha/d}, \quad v = \sqrt{\frac{2r(1-r)}{2r-1}}, \quad \eta = \frac{\sqrt{2}u}{v}, \quad \beta = 2\eta, \quad c = tu - p, \tag{19}$$

and ten real numbers

$$\begin{aligned} &y_0 = c^2, && y_1 = -2\sqrt{2}uc, && y_2 = 2\sqrt{2}c, \\ &y_3 = 2u^2 + \eta(2r - v^2), && y_4 = -2\beta t, && y_5 = \eta v^2, \\ &y_6 = -4u, && y_7 = 2, && y_8 = \beta, \qquad y_9 = 0. \end{aligned} \tag{20}$$

In the lexicographic basis $|00\rangle, |01\rangle, \ldots, |22\rangle$, let H_0 have diagonal entries $y_0, y_3, y_5, y_7, y_8, y_9$ at indices $0, 3, 4, 6, 7, 8$, respectively. Its only nonzero off-diagonal entries and their equal adjoints are

$$(H_0)_{0,4} = y_1/2, \quad (H_0)_{0,8} = y_2/2, \quad (H_0)_{3,7} = y_4/2, \quad (H_0)_{4,8} = y_6/2. \tag{21}$$

Define $W_0 = W_{H_0}$ as in (7). Its symmetric blocks on $\mathcal{K}$ are

$$\begin{aligned} &C_\zeta = 2, \qquad C_R = \begin{pmatrix} y_3 & -\beta k \\ -\beta k & 1+\eta \end{pmatrix}, \\ &C_T = nn^{\mathsf{T}} + \beta mm^{\mathsf{T}}, \qquad C_V = \eta jj^{\mathsf{T}}, \qquad C_w = 0, \\ &n = (c, -u, 1, 0)^{\mathsf{T}}, \quad m = (0, -k, 0, 1)^{\mathsf{T}}, \quad j = (v, -1)^{\mathsf{T}}. \end{aligned} \tag{22}$$

These identities follow by compression in the bases of Table 1. For example $(C_T)_{12} = y_6/4$ and $(C_V)_{01} = y_6/(2\sqrt{2})$: the factors account for the two real off-diagonal entries in a trace pairing. Exact interval bounds give $y_3 > 53$ and $334 < \det C_R < 335$. Thus C_R is positive definite and the remaining blocks are manifestly positive semidefinite. The antisymmetric blocks have identical marginal images to the corresponding symmetric principal blocks, by Section 3. Their trace pairing with H_0 is therefore identical, so they too are positive semidefinite. This proves $P_{\mathcal{K}} W_0 P_{\mathcal{K}} \succeq 0$ on all 14 support dimensions.

Complementary slackness can be checked without an eigenvalue calculation: $n^{\mathsf{T}}G = m^{\mathsf{T}}G = 0$ and $j^{\mathsf{T}}h = 0$. Consequently $q(r) = 0$, where the variable-transmissivity expectation is

$$\begin{aligned} q(\lambda) := 3\,\mathrm{Tr}(H_0 J_\lambda) = y_0 + \frac{y_1\sqrt{\lambda}}{\sqrt{2}} + \frac{y_2\lambda}{4} + y_3(1-\lambda) + y_4\sqrt{\lambda}(1-\lambda) + y_5\lambda \\ + \frac{y_6\lambda^{3/2}}{\sqrt{2}} + y_7(1-\lambda)^2 + 2y_8\lambda(1-\lambda) + y_9\lambda^2. \end{aligned} \tag{23}$$

The coefficients remain fixed in this differentiation:

$$\begin{aligned} q'(\lambda) = \frac{y_1}{2\sqrt{2}\sqrt{\lambda}} + \frac{y_2}{4} - y_3 + \frac{y_4(1-3\lambda)}{2\sqrt{\lambda}} + y_5 + \frac{3y_6\sqrt{\lambda}}{2\sqrt{2}} \\ - 2y_7(1-\lambda) + 2y_8(1-2\lambda). \end{aligned} \tag{24}$$

Rational outward interval evaluation on $[\inf I_r, 2/3]$ yields $-83 < q'(\lambda) < -81$, as detailed in Appendix A. Hence $q(\lambda) < 0$ for every $r < \lambda \le 2/3$. This statement is uniform over the interval, rather than a test of sample points.

6 Full-space witnesses and the remaining transmissivities

Positivity of W_0 outside $\mathcal{K}$ has not been assumed. Introduce

$$F = \mathrm{diag}(0,1,1,0,0,1,0,0,0). \tag{25}$$

It marks the forbidden output populations $b > a$ and satisfies $\mathrm{Tr}(FJ_\lambda) = 0$ for every λ. The associated W_F vanishes on $\mathcal{K}$ and is diagonal with entries at least $1/2$ on $\mathcal{K}^\perp$. The rational coefficient bounds give $|y_i| < 60$; the 27-dimensional row-sum bound then gives $\|W_0\|_{\mathrm{op}} < 27 \cdot 60 < K_0 := 2000$.

For $r < \lambda \le 2/3$ define

$$\epsilon_\lambda = -q(\lambda)/6 > 0, \qquad M_\lambda = 2\left(K_0 + \frac{K_0^2}{\epsilon_\lambda}\right), \qquad H_\lambda = H_0 + \epsilon_\lambda I_9 + M_\lambda F. \tag{26}$$

In $\mathcal{K} \oplus \mathcal{K}^\perp$, write W_{H_λ} in block form $\left(\begin{smallmatrix} A_* & B_* \\ B_*^\dagger & D_* \end{smallmatrix}\right)$. We have $A_* \succeq \epsilon_\lambda I$, $\|B_*\| \le K_0$, and $D_* \succeq (-K_0 + \epsilon_\lambda + M_\lambda/2)I$. Therefore

$$D_* - B_*^\dagger A_*^{-1} B_* \succeq \left(-K_0 + \epsilon_\lambda + \frac{M_\lambda}{2} - \frac{K_0^2}{\epsilon_\lambda}\right) I = \epsilon_\lambda I > 0. \tag{27}$$

Thus the full W_{H_λ} is positive definite, whereas

$$\mathrm{Tr}(H_\lambda J_\lambda) = q(\lambda)/3 + \epsilon_\lambda = q(\lambda)/6 < 0. \tag{28}$$

For any extension satisfying (4), $\mathrm{Tr}(W_{H_\lambda} X) = \mathrm{Tr}(H_\lambda J_\lambda)$ would be nonnegative, a contradiction. The penalty grows without bound as $\lambda \downarrow r$, but is finite for each strictly larger transmissivity.

Let $\mathcal{E}_\mu$ be vacuum pure loss restricted to $\mathcal{H}_3$. Its Kraus operators are

$$L_\ell(\mu) = \sum_{n=\ell}^{2} \sqrt{\binom{n}{\ell}} (1-\mu)^{\ell/2} \mu^{(n-\ell)/2} |n-\ell\rangle\langle n|, \quad \ell = 0,1,2. \tag{29}$$

They satisfy $\sum_\ell L_\ell^\dagger L_\ell = I_3$ throughout $[0,1]$. Loss and number dephasing commute, and the exact loss composition law gives

$$\mathcal{E}_\mu \circ \mathcal{N}_\lambda = \mathcal{N}_{\mu\lambda} \tag{30}$$

(Lemma 28 and Eq. (67) in Ref. [1]). It can also be checked directly on the nine matrix units using (1) and (29).

For $0 \le \lambda \le r$, set $\Psi = \mathcal{E}_{\lambda/r}$ and

$$X_\lambda = (\mathrm{id}_A \otimes \Psi \otimes \Psi)(X_r). \tag{31}$$

Complete positivity, trace preservation and applying the same map to both outputs preserve positivity, normalization and swap invariance. Its marginals are J_λ by (30).

For $2/3 \le \lambda \le 1$, take $U = 2/3$ and the operator H_U from (26). With $\Psi = \mathcal{E}_{U/\lambda}$, define

$$H_\lambda = (\mathrm{id}_A \otimes \Psi^\dagger)(H_U). \tag{32}$$

The adjoint is completely positive and unital, giving

$$\begin{aligned} W_{H_\lambda} &= (\mathrm{id}_A \otimes \Psi^\dagger \otimes \Psi^\dagger)(W_{H_U}) \succeq 0, \\ \mathrm{Tr}(H_\lambda J_\lambda) &= \mathrm{Tr}(H_U J_U) < 0. \end{aligned} \tag{33}$$

The unital identity is needed here for both tensor terms in W. Together (18), (26), (31) and (32) establish every assertion of Theorem 1, including both physical endpoints $\lambda = 0, 1$.

7 Physical consequence, comparison and limits

The exact feasible interval is $[0, r]$ for arbitrary states supported on the first three Fock levels. In particular the unassisted quantum capacity of this restriction vanishes there. The two-level threshold $2/3$ is larger by

$$2/3 - r \simeq 0.00212239863349. \tag{34}$$

By Corollary 35 of Ref. [1], non-anti-degradability of a finite restriction implies non-anti-degradability of the unrestricted channel. Thus anti-degradability of the infinite bosonic channel at this phase variance necessarily requires $\lambda \leq r$, sharpening the two-level necessary bound. Anti-degradability below r for the unrestricted channel does not follow from this restriction, and neither does positive quantum capacity above r.

The closest prior calculation is the three-level numerical boundary in Ref. [1]; the advance is exact algebraic and operator certification at one specified variance. The symmetry-class criterion of Ranade [3] concerns invariance under the full family of diagonal unitaries. We use only number-phase covariance and do not apply that criterion to the present Choi matrix. Our explicit construction and separation argument supply the threshold directly. The literature comparison supports this narrow contribution; exhaustive priority under every possible transformed representation is not asserted.

This is an exact finite-restriction result, with rational error control for the algebraic signs. It is not a general-γ formula, a capacity formula, a two-way communication boundary or an empirical claim. The support transfer and full-space penalty exhibit how a singular symmetric-extension endpoint can be certified without treating numerical solver feasibility as a proof.

A Exact arithmetic and reproducibility

The auxiliary sign checks use rational endpoints, exact polynomial arithmetic and outward bounds for square roots. For a nonnegative rational a, with $D = 10^{30}$, compute

$$n = \left\lfloor \sqrt{\lfloor aD^2 \rfloor} \right\rfloor. \tag{35}$$

Then $n/D \leq \sqrt{a} < (n+1)/D$. Apply this to the lower and upper endpoints of a nonnegative interval. Addition and multiplication use the endpoint extrema; division is performed only when the denominator interval excludes zero. All such arithmetic uses integers and rational numbers, with no floating-point tolerance.

Table 2 lists conservative rational enclosures, written with terminating decimals interpreted as exact rationals. The sign and operator conditions follow with substantial margins. The coefficient intervals in (20) are computed with $r \in I_r$, and the derivative interval uses $\lambda \in [\inf I_r, 2/3]$ independently. This enlarged interval covers every required value; no cancellation requiring numerical correlation between the two variables is assumed.

For root isolation, take the Sturm chain $S_0 = P$, $S_1 = P'$, $S_{j+1} = -\operatorname{rem}(S_{j-1}, S_j)$, stopping at the nonzero constant. Exact sign variations at the endpoints of I_r are respectively 3 and 2, so the root count is one. The same computation gives one root in $(3/5, 2/3)$ and in the interval of (8). These checks, the polynomial identity (17), endpoint marginal identities, dual block identities and the nine loss-composition units are supplied as reproducible symbolic calculations in Online Resource 1. Numerical optimization was used only to discover the candidate, not to validate it.

The recorded computational environment is Python 3.12.14, SymPy 1.14.0, NumPy 2.3.5 and SciPy 1.18.1. The decisive symbolic and rational checks require SymPy and the Python standard library; NumPy and SciPy support the optional discovery calculations. The supplementary README separates these tasks and gives regeneration commands.

Table 2: Conservative exact rational bounds sufficient for the radical branch, endpoint positivity and full-space witness construction.

Quantity	Lower bound	Upper bound
$A(r)$	0.0086	0.0087
$B(r)$	2.34	2.35
$C(r)$	2.64	2.65
$C(r) - A(r) - B(r)$	0.28	0.29
d	0.0063	0.0064
α	0.169	0.171
$2r - 1$	0.329	0.330
w	0.10	0.12
y_3	53	54
$\det C_R$	334	335
Each y_i	-60	60
$q'(\lambda)$ on $[\inf I_r, 2/3]$	-83	-81

Statements and Declarations

Data and code availability. The channel is specified analytically by (1). Own symbolic verification code, exact certificate data and reproduction instructions accompany the manuscript as Online Resource 1; no experimental dataset is used. The supplementary material does not include third-party full-text papers.

Use of generative artificial intelligence. OpenAI Codex was used in analytic candidate development, calculation and code preparation, computational checking, literature screening and manuscript drafting. The exact checks use the symbolic and rational arithmetic methods described above. This disclosure covers substantive research and text generation, beyond copy editing.

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